

Sentinel Browser — Important Literature Papers

Focused literature survey containing the 8 most relevant papers for the proposed Sentinel Browser project.

No. + Research Paper + Author(s) / Year	Research Objective	Method / Key Contribution	Major Findings	Limitation / Research Gap	What We Take for Sentinel Browser	Improvement in Our Project
1. PentestGPT: An LLM-Empowered Automatic Penetration Testing Tool G. Deng et al., 2023	To automate penetration-testing tasks using an LLM while maintaining task context.	Uses multiple interacting modules for penetration testing with iterative reasoning and information exchange.	Demonstrated substantial improvement in task completion compared with a GPT-3.5 baseline.	LLMs may lose overall context and can struggle with reliable end-to-end execution.	Modular AI-assisted security reasoning.	Sentinel separates deterministic browser/scanner execution from local LLM interpretation and verification.
2. PentestAgent: Incorporating LLM Agents to Automated Penetration Testing X. Shen et al., 2025	To improve automated penetration testing using coordinated LLM agents.	Multi-agent workflow covering intelligence gathering, vulnerability analysis and exploitation, supported by RAG.	Reported improved benchmark performance and testing efficiency.	Agent orchestration, tool use and grounding remain challenging.	Multi-stage and agent-based security reasoning.	Sentinel uses a controlled state-machine workflow with evidence-grounded local AI rather than unrestricted autonomous exploitation.
3. AWE: Adaptive Agents for Dynamic Web Penetration Testing Akshat Singh Jaswal, 2026	To develop adaptive agents for dynamic web penetration testing.	Memory-augmented multi-agent architecture with vulnerability-specific pipelines and browser verification.	Reported strong performance on the XBOW benchmark together with token-efficiency improvements.	Focuses primarily on autonomous offensive web testing.	Vulnerability-specific reasoning, memory and browser-aware verification.	Sentinel combines browser instrumentation, multiple scanners, visual evidence and local AI analysis in a defensive security lab.
4. LLM Agents Can Autonomously Hack Websites R. Fang et al., 2024	To investigate whether LLM agents can autonomously discover and exploit website vulnerabilities.	Autonomous website hacking involving reconnaissance, reasoning and exploitation.	GPT-4 demonstrated meaningful autonomous website exploitation capability.	Emphasis is on offensive autonomy rather than evidence-first defensive assessment.	Shows the feasibility of AI-assisted web-security reasoning.	Sentinel limits AI to interpretation, verification, explanation and remediation of collected evidence.
5. EnIGMA: Enhanced Interactive Generative Model Agent for CTF Challenges Talor Abramovich et al., 2024	To enable language-model agents to solve interactive CTF challenges.	Provides interactive tools such as debugger/server connections and agent tool use.	Achieved strong results across multiple CTF benchmarks.	The work identifies hallucinated observations and other agent reliability problems.	Ground agents in real tool observations.	Sentinel records real browser, network and scanner evidence before the AI produces conclusions.

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6. NAUTILUS: Automated RESTful API Vulnerability Detection G. Deng et al., 2023	To improve automated vulnerability detection in REST APIs.	Models API state and operation relationships and generates valid interaction sequences.	Reported higher vulnerability discovery and API-operation coverage, including real-world vulnerabilities.	Focused mainly on REST API testing rather than complete browser-security assessment.	State-aware application interaction and sequence generation.	Sentinel uses Playwright-driven exploration to maintain browser state and collect evidence across real user flows.
7. Reflexion: Language Agents with Verbal Reinforcement Learning Noah Shinn et al., 2023	To improve language-agent performance through verbal feedback and memory.	Uses episodic memory and verbal reinforcement without changing model weights.	Demonstrated significant improvements across several reasoning and agent tasks.	Reflection can reinforce incorrect assumptions when feedback is not grounded.	Memory, reflection and iterative analysis.	Sentinel uses checkpoints and event logs so AI conclusions can be traced back to concrete security observations.
8. PenHeal: A Two-Stage LLM Framework for Automated Pentesting and Optimal Remediation J. Huang & Q. Zhu, 2024	To combine automated penetration testing with vulnerability remediation.	Two-stage Pentest and Remediation modules using prompting techniques and external security knowledge.	Reported improvements in vulnerability coverage, remediation effectiveness and cost efficiency.	Requires reliable grounding and is strongly oriented toward automated pentesting.	Connect vulnerability detection with remediation guidance.	Sentinel adds local OWASP RAG, evidence-backed explanations, visual evidence and replayable browser workflows.

Why These 8 Papers Were Selected

Paper	Relevance to Sentinel Browser
PentestGPT	Foundational LLM-based penetration-testing work.
PentestAgent	Directly relevant to multi-agent security automation and RAG.
AWE	Most directly aligned with dynamic web penetration testing and browser verification.
LLM Agents Can Autonomously Hack Websites	Provides direct evidence of LLM-based autonomous web security capabilities.
EnIGMA	Important for interactive tool use and the problem of hallucinated observations.
NAUTILUS	Relevant to state-aware web/API interaction and vulnerability detection.
Reflexion	Provides the basis for memory, feedback and iterative agent improvement.
PenHeal	Directly connects security testing with remediation, matching Sentinel's AI remediation goal.

Recommended literature-review flow: PentestGPT → PentestAgent → AWE → Autonomous Web Hacking → EnIGMA → NAUTILUS → Reflexion → PenHeal. Together these papers cover LLM-based penetration testing, web security, agents, browser/tool interaction, state-aware testing, evidence grounding, memory/reflection and remediation.

Proposed Research Gap

Existing LLM-based penetration-testing research predominantly focuses on autonomous attack planning, exploitation, or specialized vulnerability discovery, whereas there is comparatively less emphasis on a locally executable browser-security assessment system that combines deterministic browser instrumentation, multiple security scanners, evidence-grounded LLM analysis, visual evidence, replayability, and real-time browser-extension integration.

Sentinel Browser positioning: Existing LLM-based penetration-testing research primarily asks the LLM to perform the security testing; Sentinel Browser uses deterministic browser instrumentation to perform and record observations, and uses a local LLM to interpret, verify, explain and remediate those observations.